

Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the **mean, median, and mode**.
2. Find the **range**.
3. Calculate the **variance and standard deviation**.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Solution:

1. Find the **mean, median, and mode**.

Formula:

Mean = (Sum of all observations) / n

Substitution:

Mean = $(4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30) / 9$

Mean = $82 / 9$

Mean = 9.11

Formula:

Median = $(n + 1) / 2$ th observation

Ordered Data:

4, 5, 5, 6, 7, 8, 8, 9, 30

Median = 5th observation

Median = 7

Mode = 5 and 8

Answer:

$$\text{Mean} = 9.11$$

$$\text{Median} = 7$$

$$\text{Mode} = 5 \text{ and } 8$$

2. Find the **range**

Formula:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Substitution:

$$\text{Range} = 30 - 4$$

$$\text{Range} = 26$$

Answer:

$$\text{Range} = 26$$

3. Calculate the **variance and standard deviation**.

Formula:

$$\text{Variance} = \Sigma(X - \text{Mean})^2 / n$$

X	X – 9.11	(X – 9.11)²
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

$$\Sigma(X - \text{Mean})^2 = 512.89$$

Substitution:

$$\text{Variance} = 512.89 / 9$$

$$\text{Variance} = 56.99$$

Formula:

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Substitution:

$$\text{Standard Deviation} = \sqrt{56.99}$$

$$\text{Standard Deviation} = 7.55$$

Answer:

$$\text{Variance} = 56.99$$

$$\text{Standard Deviation} = 7.55$$

4. Identify the outlier

$$Q1 = 5$$

$$Q3 = 8$$

$$\text{IQR} = 8 - 5 = 3$$

$$\text{Lower Limit} = 5 - (1.5 \times 3) = 0.5$$

$$\text{Upper Limit} = 8 + (1.5 \times 3) = 12.5$$

Since $30 > 12.5$, it is an outlier.

Answer:

$$\text{Outlier} = 30$$

5. Which measure of central tendency best represents the typical salary? Explain why

The salary of 30 lakhs is an outlier, which increases the mean. The median is not affected by this extreme value.

Answer:

Median is the best measure of central tendency because it is not affected by the outlier and better represents the typical employee salary.

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Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be **right-skewed or left-skewed**.
5. Explain why the median may be more useful than the mean.

Solution:

1. Calculate the mean and median.

Formula:

Mean = (Sum of all observations) / n

Substitution:

Mean = (120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300) / 10

Mean = 1505 / 10

Mean = 150.5 ms

Formula:

Median = (5th value + 6th value) / 2

Ordered Data:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Substitution:

Median = (132 + 135) / 2

Median = 267 / 2

Median = 133.5 ms

Answer:

Mean = 150.5 ms

Median = 133.5 ms

2. Find the range.

Formula:

Range = Maximum Value – Minimum Value

Substitution:

Range = 300 – 120

Range = 180 ms

Answer:

Range = 180 ms

3. Calculate the variance and standard deviation.

Formula:

Variance = $\Sigma(X - \text{Mean})^2 / n$

X	X – 150.5	(X – 150.5) ²
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

$$\Sigma(X - \text{Mean})^2 = 25580.50$$

Substitution:

$$\text{Variance} = 25580.50 / 10$$

$$\text{Variance} = 2558.05$$

Formula:

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Substitution:

$$\text{Standard Deviation} = \sqrt{2558.05}$$

$$\text{Standard Deviation} = 50.58 \text{ ms}$$

Answer:

$$\text{Variance} = 2558.05$$

$$\text{Standard Deviation} = 50.58 \text{ ms}$$

4. Determine whether the data is likely to be right-skewed or left-skewed

The response time of **300 ms** is much larger than the remaining values. This causes the tail of the distribution to extend towards the right.

Answer:

The data is **right-skewed (positively skewed)**.

5. . Explain why the median may be more useful than the mean.

The mean is affected by the unusually high response time of **300 ms**, while the median is not.

Answer:

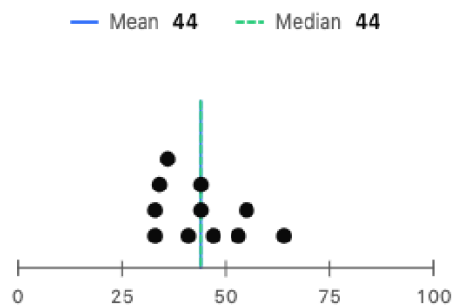
The **median** is more useful because it is not affected by the extreme value and better represents the typical website response time.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

- 1. Find Q1, Q2 and Q3.
- 2. Calculate the **IQR**.
- 3. Use the **1.5 × IQR rule** to identify any outliers.
- 4. Find the mean and median.
- 5. Explain how the outlier affects the mean.

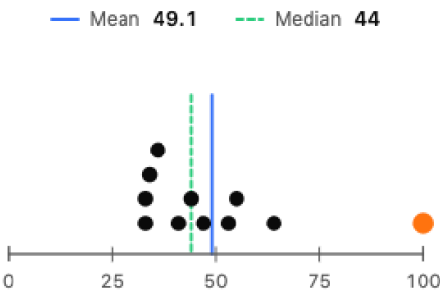


No outlier: mean is 44; median is 44

No outlier

High outlier

New data set



Outlier: mean moves 44 → 49.1; median stays 44

No outlier

High outlier

New data set

Solution:

1.

Ordered Data:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Q1 = Median of first 6 values

$$Q1 = (48 + 50) / 2$$

$$Q1 = 98 / 2$$

$$Q1 = \mathbf{49}$$

$$Q2 \text{ (Median)} = (52 + 55) / 2$$

$$Q2 = 107 / 2$$

$$Q2 = \mathbf{53.5}$$

Q3 = Median of last 6 values

$$Q3 = (60 + 62) / 2$$

$$Q3 = 122 / 2$$

$$Q3 = \mathbf{61}$$

Answer:

$$Q1 = \mathbf{49}$$

$$Q2 = \mathbf{53.5}$$

$$Q3 = \mathbf{61}$$

2.

Formula:

$$\mathbf{IQR} = Q3 - Q1$$

Substitution:

$$IQR = 61 - 49$$

$$IQR = \mathbf{12}$$

Answer:

$$IQR = \mathbf{12}$$

3.

Formula:

$$\text{Lower Limit} = Q1 - (1.5 \times \text{IQR})$$

$$\text{Upper Limit} = Q3 + (1.5 \times \text{IQR})$$

Substitution:

$$1.5 \times \text{IQR} = 1.5 \times 12 = 18$$

$$\text{Lower Limit} = 49 - 18 = 31$$

$$\text{Upper Limit} = 61 + 18 = 79$$

Values below 31 or above 79 are outliers.

Since **95** > **79**, it is an outlier.

Answer:

Outlier = **95**

4.

Formula:

$$\text{Mean} = (\text{Sum of all observations}) / n$$

Substitution:

$$\text{Mean} = (42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95) / 12$$

$$\text{Mean} = 684 / 12$$

$$\text{Mean} = \mathbf{57}$$

Formula:

$$\text{Median} = (6\text{th value} + 7\text{th value}) / 2$$

Substitution:

$$\text{Median} = (52 + 55) / 2$$

$$\text{Median} = 107 / 2$$

$$\text{Median} = \mathbf{53.5}$$

Answer:

$$\text{Mean} = \mathbf{57}$$

$$\text{Median} = \mathbf{53.5}$$

5.

The value **95** is an outlier. It increases the mean considerably, while the median changes very little.

Answer:

The outlier **95** increases the mean but has very little effect on the median. Therefore, the median is a better measure of the typical student performance.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Solution:

1.

Formula:

Mean = (Sum of all observations) / n

Substitution:

Mean = $(18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35) / 10$

Mean = $250 / 10$

Mean = **25**

Formula:

Median = (5th value + 6th value) / 2

Ordered Data:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Substitution:

Median = $(24 + 25) / 2$

Median = $49 / 2$

Median = **24.5**

Mode = **22**

Answer:

Mean = **25**

Median = **24.5**

Mode = **22**

2.

Formula:

$$\text{Variance} = \Sigma(X - \text{Mean})^2 / n$$

X	X – 25	(X – 25)²
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

$$\Sigma(X - \text{Mean})^2 = 228$$

Substitution:

$$\text{Variance} = 228 / 10$$

$$\text{Variance} = \mathbf{22.8}$$

Formula:

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Substitution:

$$\text{Standard Deviation} = \sqrt{22.8}$$

Standard Deviation = **4.77**

Answer:

Variance = **22.8**

Standard Deviation = **4.77**

3.

Formula:

Coefficient of Variation (CV) = (Standard Deviation / Mean) $\times$ 100

Substitution:

$CV = (4.77 / 25) \times 100$

$CV = 0.1908 \times 100$

CV = 19.08%

Answer:

Coefficient of Variation = **19.08%**

4.

The standard deviation measures how much the daily orders differ from the average number of orders.

Since the standard deviation is **4.77**, the daily orders generally vary by about **4.77 orders** from the mean of **25 orders**.

Answer:

The standard deviation tells the company how consistent the daily orders are. A lower standard deviation indicates more consistent daily orders, while a higher standard deviation indicates greater variation in the number of orders.

5.

The coefficient of variation is **19.08%**, which indicates a moderate level of variability.

Answer:

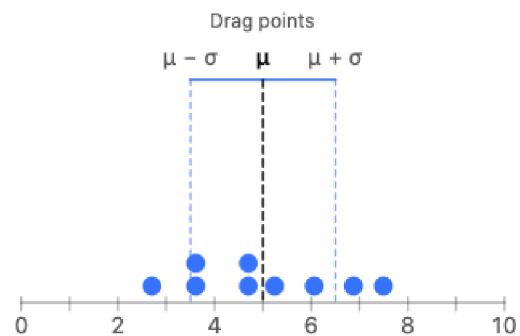
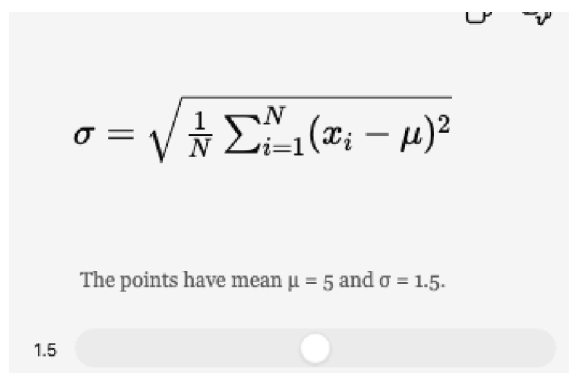
The order data is **moderately variable** because the variation in daily orders is not too high compared to the average number of orders.

Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6 **Model B:** 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.



Solution:

1.

Formula:

Mean = (Sum of all observations) / n

Model A

Substitution:

Mean = $(2 + 3 + 3 + 4 + 4 + 5 + 5 + 6) / 8$

Mean = $32 / 8$

Mean = **4**

Model B

Substitution:

Mean = $(1 + 1 + 2 + 4 + 5 + 7 + 8 + 9) / 8$

$$\text{Mean} = 37 / 8$$

$$\text{Mean} = \mathbf{4.625}$$

Answer:

$$\text{Mean of Model A} = \mathbf{4}$$

$$\text{Mean of Model B} = \mathbf{4.625}$$

2.

Formula:

$$\mathbf{\text{Variance}} = \Sigma(X - \text{Mean})^2 / n$$

Model A

X	X - 4	(X - 4)²
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

$$\Sigma(X - \text{Mean})^2 = 12$$

Substitution:

$$\text{Variance} = 12 / 8$$

$$\text{Variance} = \mathbf{1.5}$$

Model B

X	X - 4.625	(X - 4.625)²
1	-3.625	13.14
1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14

X	X – 4.625	(X – 4.625)²
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

$\Sigma(X - \text{Mean})^2 = 69.88$

Substitution:

Variance = 69.88 / 8

Variance = **8.73**

Answer:

Variance of Model A = **1.5**

Variance of Model B = **8.73**

3.

Formula:

Standard Deviation = $\sqrt{\text{Variance}}$

Model A

Substitution:

Standard Deviation = $\sqrt{1.5}$

Standard Deviation = **1.22**

Model B

Substitution:

Standard Deviation = $\sqrt{8.73}$

Standard Deviation = **2.95**

Answer:

Standard Deviation of Model A = **1.22**

Standard Deviation of Model B = **2.95**

4.

Model A has a smaller standard deviation than Model B.

Answer:

Model A has more consistent predictions because its prediction errors are closer to the mean and have a lower standard deviation.

5.

A lower mean error alone is not enough to judge a model because it does not show how much the errors vary.

Answer:

No. A model with a lower mean error cannot always be considered better. Variance and standard deviation should also be considered because they indicate the consistency of the model's predictions.